CHAPTER EXERCISE: Answers for this chapter start on page 307.

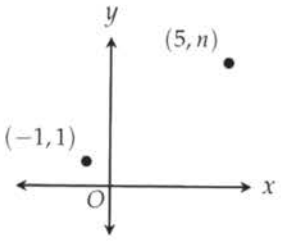
A calculator should NOT be used on the following questions.

1

What is the equation of the line parallel to the y-axis and 3 units to the right of the y-axis?

A) $x = -3$
B) $x = 3$
C) $y = -3$
D) $y = 3$

2



Note: Figure not drawn to scale.

In the figure above, the slope of the line through the two plotted points is $\frac{1}{3}$. What is the value of n?

A) 9
B) 4
C) 3
D) $\frac{7}{3}$

3

In the xy-plane, a line has an x-intercept of -2 and a y-intercept of -4. What is the slope of the line?

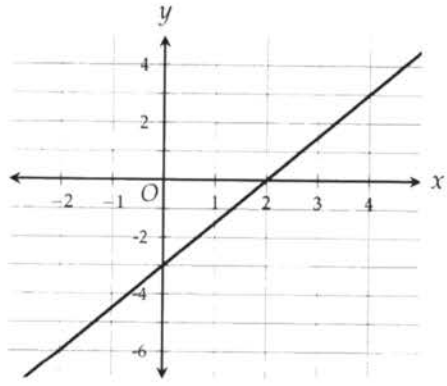
A) -2
B) $-\frac{1}{2}$
C) $\frac{1}{2}$
D) 2

4

In the xy-plane, points $(-3,5)$ and $(6,8)$ lie on line l. Which of the following points is also on line l?

A) $(0,6)$
B) $(3,8)$
C) $(9,10)$
D) $(12,11)$

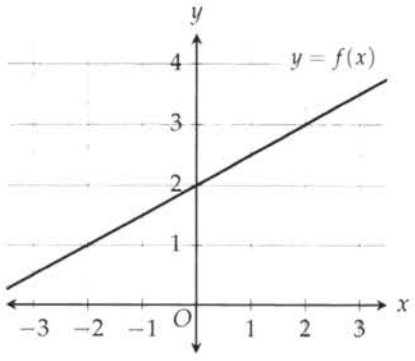
5



The graph of line l is shown in the xy-plane above. Which of the following is an equation of a line that is parallel to line l?

A) $y = -\frac{2}{3}x + 2$
B) $y = \frac{2}{3}x + 10$
C) $y = \frac{3}{2}x - 4$
D) $y = 3x - 1$

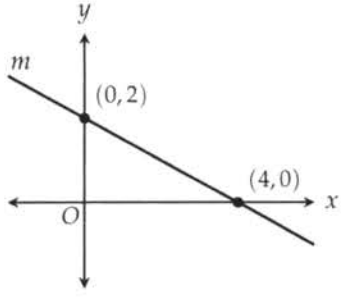
6



In the xy-plane above, the graph of the linear function f is perpendicular to the graph of the linear function g (not shown). If the graphs of f and g intersect at the point $\left(1, \frac{5}{2}\right)$, what is the value of $g(-1)$?

A calculator is allowed on the following questions.

7



What is the slope of the line m in the figure above?

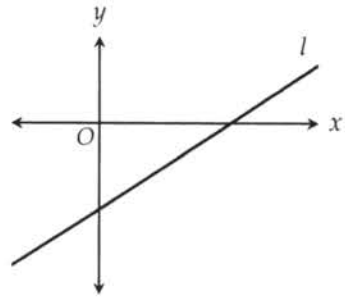
A) -2

B) $-\frac{1}{2}$

C) $\frac{1}{4}$

D) $\frac{1}{2}$

8



Line l in the xy-coordinate system above can be represented by the equation $y = mx + b$. Which of the following must be true?

A) $mb > 0$

B) $mb < 0$

C) $mb = 0$

D) $mb = 1$

9

The line $y = -2x - 2$ is perpendicular to line l. If these two lines have the same y-intercept, which of the following could be the equation of line l ?

A) $y = -2x - 2$

B) $y = 2x - 2$

C) $y = -\frac{1}{2}x - 2$

D) $y = \frac{1}{2}x - 2$

10

The slope of line l is $\frac{1}{2}$ and its y-intercept is 3. What is the equation of the line perpendicular to line l that goes through $(1,5)$?

A) $y = -2x + 3$

B) $y = -2x + 7$

C) $y = -\frac{1}{2}x + \frac{11}{2}$

D) $y = \frac{1}{2}x + \frac{9}{2}$

11

A line with a slope of $\frac{2}{3}$ passes through the points $(1,4)$ and $(x,10)$. What is the value of x ?

A) 4

B) 6

C) 8

D) 10

12

Day	Average speed, s (miles per hour)	Number of calories burned, c
Monday	7.2	616
Thursday	6.8	584
Friday	7.9	672
Saturday	8.5	720

On certain days of the week, Elaine runs for an hour on a treadmill. For each day that she ran in the last week, the table above shows the average speed s at which she ran, in miles per hour, and the number of calories c she burned during the run. If the relationship between c and s can be modeled by a linear function, which of the following functions best models the relationship?

A) $c(s) = 30s + 400$

B) $c(s) = 60s + 210$

C) $c(s) = 80s + 40$

D) $c(s) = 90s - 30$

13

If m and b are real numbers and $m > 0$ and $b > 0$, then the line whose equation is $y = mx + b$ cannot contain which of the following points?

A) $(0,1)$

B) $(1,1)$

C) $(-1,1)$

D) $(0,-1)$

14

In the xy-plane, the line with equation $ax - \frac{1}{3}y = 8$, where a is a constant, passes through the point $(2,6)$. What is the x-coordinate of the x-intercept of the line?

15

$$y = \frac{a}{b}x + c$$

$$y = \frac{d}{e}x + c$$

The equations of two perpendicular lines in the xy-plane are shown above, where a, b, c, d, and e are constants. If $0 < \frac{a}{b} < 1$, which of the following must be true?

A) $\frac{d}{e} < -1$

B) $-1 < \frac{d}{e} < 0$

C) $0 < \frac{d}{e} < 1$

D) $\frac{d}{e} > 1$

Chapter 14: Lines

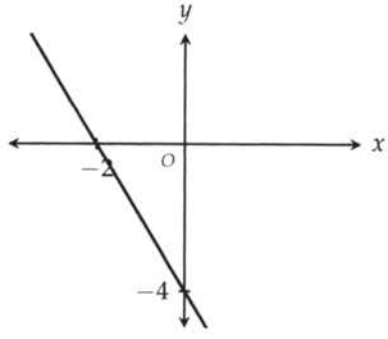
CHAPTER EXERCISE:

1. [B] A vertical line that intersects the x-axis at 3 has an equation of $x = 3$.

2. [C]

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{1}{3}$$
$$\frac{n-1}{5-(-1)} = \frac{1}{3}$$
$$\frac{n-1}{6} = \frac{1}{3}$$
$$n - 1 = 2$$
$$n = 3$$

3. [A] Draw a line from the x-intercept of -2 to the y-intercept of -4.



As you can see, it goes 4 units down for every 2 units to the right. The slope is $\frac{-4}{2} = -2$.

4. [A] The slope of line l is $\frac{8-5}{6-(-3)} = \frac{3}{9} = \frac{1}{3}$. Using point-slope form,

$$y - y_1 = m(x - x_1)$$
$$y - 8 = \frac{1}{3}(x - 6)$$
$$y = \frac{1}{3}x + 6$$

At this point, we test each answer choice by plugging in the x-coordinate and verifying the y-coordinate. Only answer (A) works.

5. [C] The graph of line l goes up three units for every two units to the right, which means its slope is $\frac{3}{2}$. A parallel line must have the same slope. Only answer choice (C) gives an equation of a line with the same slope.

6. [6.5 or $\frac{13}{2}$] From the graph, we can see that f goes up 1 unit for every 2 units to the right, which means its slope is $\frac{1}{2}$. Since g is perpendicular to f, the slope of g must be -2 (the negative reciprocal). Since g passes through the point $\left(1, \frac{5}{2}\right)$, we can use point-slope form to find the equation of g:

$$y - y_1 = m(x - x_1)$$
$$y - \frac{5}{2} = -2(x - 1)$$
$$y = -2x + 2 + \frac{5}{2}$$
$$y = -2x + \frac{9}{2}$$

Finally, $g(-1) = -2(-1) + \frac{9}{2} = 2 + 4\frac{1}{2} = 6\frac{1}{2}$. A quicker way would've been to work backwards from the point $\left(1, \frac{5}{2}\right)$, knowing that the slope is -2. So on the graph of g, 1 unit to the left brings us to $\left(0, \frac{5}{2} + 2\right) = \left(0, 4\frac{1}{2}\right)$, and 1 more unit to the left brings us to $\left(-1, 4\frac{1}{2} + 2\right) = \left(-1, 6\frac{1}{2}\right)$. Therefore, $g(-1) = 6.5$.

7. [B]

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{2-0}{0-4} = \frac{2}{-4} = -\frac{1}{2}$$

8. [B] From the graph, slope m is positive and y-intercept b is negative. Therefore, $mb < 0$.

9. $\boxed{D}$ The line $y = -2x - 2$ has a slope of -2 and a y-intercept of -2. Line l must have a slope that is the negative reciprocal of -2, which is $\frac{1}{2}$. Since they have the same y-intercept, the equation of line l must be $y = \frac{1}{2}x - 2$.

10. $\boxed{B}$ The line we're looking for must have a slope that is the negative reciprocal of $\frac{1}{2}$, which is -2.

$$y = -2x + b$$

Plugging in the point $(1,5)$,

$$5 = -2(1) + b$$

$$7 = b$$

Now that we have b, the line is $y = -2x + 7$.

11. $\boxed{D}$

$$\frac{10-4}{x-1} = \frac{2}{3}$$

$$\frac{6}{x-1} = \frac{2}{3}$$

Cross multiplying,

$$\begin{aligned} 2(x-1) &= 18 \\ 2x - 2 &= 18 \\ 2x &= 20 \\ x &= 10 \end{aligned}$$

12. $\boxed{C}$ We can use the values from any two days to find the line that best models the data. Let's use the values from Monday and Thursday to calculate the slope: $\frac{616-584}{7.2-6.8} = 80$. At this point, we can tell the answer is probably going to be choice C, but let's find the y-intercept just to be sure. Currently, we have $c = 80s + b$. Plugging in $s = 7.2$ and $c = 616$ from Monday, we get $616 = 80(7.2) + b$, which gives $b = 616 - 80(7.2) = 40$. Therefore, $c(s) = 80s + 40$.

An alternative solution is to test each of the answer choices by plugging in the values from the table. Answer A works for Monday ($c(7.2) = 30(7.2) + 400 = 616$) but not for any of the other days. Answer B works for Saturday ($c(8.5) = 60(8.5) + 210 = 720$) but not for any of the other days. Answer D does not give the correct value of c for any of the given values of s. Only answer C gives the correct value of c for each of the given values of s. These types of questions require you to be thorough. Don't just test one case and choose the first thing that "works." You have to evaluate all the answer choices.

13. $\boxed{D}$ A line with a positive y-intercept will not cross the y-axis at a negative point. Therefore, when x is 0, y cannot be negative, which makes (E) the answer.

14. $\boxed{1.6 \text{ or } \frac{8}{5}}$ First, plug the point $(2,6)$ into the equation of the line so that we can solve for a:

$$\begin{aligned} a(2) - \frac{1}{3}(6) &= 8 \\ 2a - 2 &= 8 \\ 2a &= 10 \\ a &= 5 \end{aligned}$$

So the equation of the line is $5x - \frac{1}{3}y = 8$. The x-intercept always has a y-coordinate of 0, so if we plug in 0 for y, we get $5x = 8$, which gives $x = \frac{8}{5} = 1.6$.

15. $\boxed{A}$ One easy way to approach this problem is to make up numbers for a and b. Let $a = 1$ and $b = 2$ so that $\frac{a}{b} = \frac{1}{2}$. Since the second line is perpendicular to the first, $\frac{d}{e} = -2$, which satisfies the condition in answer choice (A).